

Article

Detection of Depression Severity in Social Media Text Using Transformer-Based Models

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Abstract: Depression, a serious mental health disorder, requires accurate classification for effective intervention. Existing methods often fail to capture nuanced emotional and linguistic cues, leading to suboptimal classification of depression severity. This study bridges this gap by leveraging content-based approaches (N-grams) and context-based methods (Sentence Transformers), alongside advanced transformer-based models, to classify mild, moderate, and severe depression using text data sourced from Reddit. By demonstrating the effectiveness of modern NLP techniques in capturing subtle contextual variations, this research highlights the potential of transformer-based models to enhance depression severity detection. The proposed framework offers a scalable and adaptable solution for real-world mental health diagnostics and early intervention systems.

Keywords: natural language processing; sentence transformers; machine learning



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1. Introduction

Depression is a complex psychological condition that affects emotions, thinking patterns, and everyday function. Whereas temporary sadness or a difficult phase is usually time-limited, depression entails a sustained period of deep sorrow, a dramatic decrease in interest or pleasure in previously enjoyed activities, and a range of chronic physical and emotional symptoms. Symptoms differ, however, by type but often involve persistent sadness, loss of interest or pleasure in most or all things, as well as lack of energy as an altered appetite or weight changes [1]. It has an impact on relationships, school, and work, and it can happen to anyone, especially people who have been affected by abuse or profound loss. Confines of the World Health Organization (WHO) [2] also predicts that approximately 3.8% of the total population suffers from depression, with 5% of adults and 5.7% of the elderly.

Addressing the pervasive impact of depression through innovative strategies for integrated diagnosis and treatment has become increasingly significant. This emphasis has further accelerated the implementation of mental health promotion programs aimed at the detection, treatment, and prevention of depression [3]. However, the public may take more time to obtain an immediate diagnosis or treatment due to self-denial and poor problem recognition [4]. Reversely, an increase in younger age depression prevalence has begun to occur. People suffering from depression are more likely to self-harm and attempt suicide than people without the disease, which may pose a severe threat to their lives as well as even to their family members. Moreover, combatting and mitigating the impact of

depression has become the most salient regional goal concerning mental health in many countries [5].

Patients are usually asked to fill out pertinent questionnaires or seek help from medical specialists in the traditional method of diagnosing depression. Nevertheless, the efficacy of these techniques is restricted because many people with depression may choose to hide their symptoms purposefully and are reluctant to share their heartfelt thoughts with others. Furthermore, several studies have used physiological markers to identify people who are mentally ill or who are considering suicide. Nevertheless, these procedures are laborious and require individuals to wear specific equipment. Additionally, they could worsen the patient's condition and develop resistance in the patients [6].

Individuals who suffer from stress and depression utilize social media sites like Reddit and Twitter to share their ideas, experiences, and other information with people using tweets and comments. As a result, social media is a valuable information source where language patterns of posts by depressed users may be examined. An emerging body of research suggests that natural language processing (NLP) methods can accurately recognize distressed individuals on social networking sites. The majority of depression detection methods now in use analyze user-posted textual material to determine the user's mental wellness. Machine learning-based identification of depression is one of those techniques that primarily uses feature engineering. Transforming unprocessed data into features that the model can comprehend more readily increases the predicted accuracy of the model [7].

Researchers have recently shown growing interest in the emerging field of automatic depression detection through social media posts. Building on previous efforts to identify depression simply, recent studies have expanded to assess the severity of depression, with a focus on predicting potential future mental health issues for social media users [8]. This study employs content-based methods, utilizing N-gram and context-based methods, leveraging sentence transformers (ST), to detect depression using data from Reddit. These approaches are integrated with classical machine learning (ML) algorithms to explore their effectiveness. Additionally, advanced transformer-based architectures are implemented end-to-end, enabling a comprehensive performance evaluation.

In contrast, this study utilizes sentence-transformer models to classify depression severity levels into four categories, allowing it to capture finer granularity than binary classification approaches. The proposed framework further simplifies the analysis by examining only three severity levels (minimum, moderate, and severe), allowing for a more precise comprehension of depressive states that permits clinical intervention. To that end, this work offers a valuable addition to the mental health diagnostic comparison space, as contextual embeddings tackle common limitations of shallow machine-learning, such as superficial semantic representation and feature engineering reliance.

This paper proceeds as follows. Section 2 provides a comprehensive review of the recent literature in this domain, highlighting advancements and ongoing challenges. Section 3 details the Gap analysis and research goals. Section 4 presents the methodological framework developed to detect depression severity levels using Reddit data. Section 5 covers the experimental setup, outcomes, and a thorough analysis of the results achieved through the proposed models. Finally, Section 6 concludes the paper, offering insights into potential avenues for future research and model refinement.

2. Literature Review

Interest in mental health has recently been a popular analysis topic, due to advancements in text mining and, more broadly, NLP and the popularity of different social media platforms. Reddit or Twitter are other significant sources of online textual data that reflect users' emotions, mental status, and experiences [9,10]. Researchers have leveraged this

information to develop automated detection systems for psychological disorders, such as stress, anxiety, and depression [11]. These systems act as fast and low-cost instruments complementing conventional approaches to mental health screening [12], such as detecting those potentially at risk, and promptly enabling them to receive support, while also contributing to mental health investigations.

The initial detection stage focused on formulating the task as a binary classification, either depressed or non-depressed [13]. This is a somewhat practical approach, but it simplifies a complicated disease. Depression does not produce uniform diagnosing levels, as One identifies the depression level as either Mild, Moderate, or Severe at a single place then it has a node problem in the detection system. Multi-class classification frameworks have been recently applied to this problem domain, combining a finer-grained understanding of depressive states and their subtle trajectories [14,15].

The early applications involved manual feature engineering. Another example of applying a similar technique of feature extraction on the textual features like part-of-speech tagging, emotive vocabularies, and polarity of the “words” of the users from social media posts [16]. Add behavioral characteristics, such as post frequency, and activity logs [17]. These features were then fed to flat machine learning classifiers such as Logistic Regression (LR), Support Vector Machine (SVM), or Decision Trees (DTs) [18]. These approaches were successful but fundamentally context and natural language (NL) semantics illiterate [19].

With the emergence of deep learning, depression detection was changed from a manual feature extraction method in traditional approaches to automated approaches that easily model complex patterns in text [20]. Models like Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) have become widespread due to their ability to encode spatial and temporal information. Techniques such as word embeddings (word2vec, fastText) improved the semantic context of the text [21], and also hybrid models (CNN-LSTM) increased classification performance for slightly different datasets.

Attention mechanisms further improved deep learning methods, giving more weight to the most relevant text elements. Methods like the Bidirectional LSTMs (BiLSTM) with attention layers also identify different levels of depression and reformulate the problem as a multi-class classification task for example, Bidirectional Encoder Representations from Transformers (BERT) [22] has considered context dependence and more nuanced differences between emotions in text.

In recent years, explainability has become a pressing issue. Although black-box models are extremely efficient for some tasks, there is a need for interpreting such models in critical domains like mental health analysis to identify potential instances of model bias and misuse. The use of Shapley Additive explanations (SHAP) and Hierarchical Attention Networks (HAN) to explain model predictions has elicited model interpretability and reliability [23]. These methods help to adopt AI-driven solutions in the mental health field [24] by allowing all involved to understand the reasoning behind a classification decision.

Ensemble techniques have also been of interest to researchers for depression detection. Ensemble methods improve robustness and accuracy in learning [25] by combining multiple models. They jointly apply stacking ensembles to aggregate binary classifiers (as outputs) for multi-label classification and report gains compared to baseline methods. Combining different architectures to learn complex patterns from mental health data can lead to high performance, as either evidence for validation [26] or simply for example.

Since these affective states are rooted in different modes of expression, multimodal approaches provide another layer for depression detection by incorporating text alongside other modalities that describe the same experience (e.g., visual and acoustic). Holistic Heatmaps used biGRU with attention layers over CNN units’ audio, video, and text data

output. In this regard, these multimodal systems show how fusing heterogeneous sources of information can increase performance and robustness in detection [27].

Moreover, Transformer-based models are a game-changer in the sophistication that we now have for depression detection models. For example, BERT and RoBERTa require contextual embedding to represent input words preprocess massive context texts, which achieve better expressiveness to gaze fine-grained language patterns [22,28]. Hybrid approaches combine models pre-trained on transformers like RoBERTa and preface them with a sequential model afterward [29], such as RoBERTa-BiLSTM that leverage both architectures to achieve SOTA results in depression severity detection.

Alternatively, researchers attempted to augment transformer-based approaches with more sophisticated feature engineering methods. For example, BERT-RF frameworks combine BERT embeddings with conventional classifiers such as Random Forests (RFs) and LR. In these approaches, the complementarity of the two families is exploited to obtain solid performance on benchmark datasets [30].

Though considerably relevant under the scope of social media and mental health, depression detection has yet to become a challenge that can effectively benefit from privacy-preserving techniques. Federated learning overcomes privacy issues for model generalizing by allowing model training to be conducted collaboratively without parties sharing raw data [31]. Such an approach is particularly salient in sensitive areas such as mental health, where users' privacy may be of utmost importance.

Transformer-based NLP models like BERT (Bidirectional Encoder Representations from Transformers) and RoBERTa (Robustly Optimized BERT Pretraining Approach) have also proven to be remarkably adept at extracting nuanced emotional cues from text data [32]. These models exploit self-attention mechanisms to dynamically capture the context at different levels, thus making them suitable for emotional and psychological analysis tasks. Also, Refs. [32,33] urge the adoption of methods such as dimensional sentiment analysis, e.g., valence-arousal modeling, to improve the interpretability of mental health diagnostics. Also, using these methods together could give a fuller depiction of emotions, which may help in more accurately detecting the severity of depression.

Even explainable LLMs are now present in the domain. While LLMs provide interpretable predictions via simulating symptoms based on diagnostic questionnaires to detect depression [34]. With these models, we can combine multiple queries into one analysis so that overall user behaviors at a fine granularity are incorporated.

The introduction of multimodal fusion techniques has also improved depression detection systems; for example, time-aware attention-based networks and a combination of temporal, acoustic, and visual data to model holistic user. These approaches highlight the importance of data fusion towards a uni-modal perspective on mental health [35]. Based on the studies in this section, we summarized the related research work in Table 1.

Based on this progress, we use the text data to propose an innovative approach for depressive symptoms analysis. Our method focuses on the preprocessing steps (tokenization, normalization, and stemming) to achieve a quality data set through the dependent use of classical machine learning algorithms for classification on transformer-based embeddings of sentences calibrated with their context [36].

Table 1. Summary analysis of reviewed literature.

Ref.	Year	Dataset	Technique	Results
[37]	2024	Reddit & Twitter	FastText Convolution Neural Network with Long Short-Term Memory (FCL)	Accuracy = 87% & 88%
[38]	2022	Sina Weibo	Ensemble Learning	Accuracy = 90.27%
[39]	2022	Reddit	Ensemble Learning	Accuracy = 75%
[40]	2022	Reddit	Deep Learning	F1-score = 0.79
[41]	2021	Twitter	Deep Learning	Accuracy = 70%
[42]	2021	Reddit	Context-based Features	Accuracy = 90%

2.1. Related Work Review and Gap Analysis

Existing research in detecting depression from social media data often relies on traditional feature extraction techniques and shallow machine learning classifiers, requiring significant domain expertise and frequently encountering issues with generalization to novel data. Some studies have explored more advanced architectures like convolutional neural networks (CNNs), bidirectional long short-term memory networks (BiLSTMs), hybrid models, and ensemble approaches. However, these models struggle to capture the nuanced, context-sensitive language specific to mental health discourse, especially in informal, user-generated text.

The recent advancement of sentence transformer models offers promising capabilities for depression detection. These models utilize Siamese and triplet neural network architectures to create semantically rich sentence embeddings, capturing deeper contextual information than traditional methods. However, despite the potential of these pre-trained models, few studies have thoroughly investigated their use in generating contextual embeddings specifically for mental health applications.

In-dimensional sentiment analysis methods like the Valence-Arousal (VA) model offer a more nuanced representation of emotional states, projecting text into a 2D space of valence (positive/negative) and arousal (intensity). This methodology has been used in mental health document retrieval to polish emotional signals and increase retrieval accuracy. For example, by integrating VA scores into the underlying word embeddings, these can be fine-tuned for more nuanced representations of user-generated text, which is especially important for detecting depression states. Future versions of this work might incorporate VA analysis using transformer-based models of representation for the interplay of sentiment dimensions and possibly improve on depression severity classification.

We addressed this gap by employing a comprehensive approach that integrates classical machine-learning techniques and transformer-based models. Initially, we utilized content-based methods, such as N-gram, and context-based methods, leveraging Sentence Transformers in combination with classical machine learning algorithms. These approaches laid the foundation for robust feature extraction and modeling. Building on this, we developed a metric using three advanced transformer-based models: BERT-base-uncased, RoBERTa-base, and Microsoft/DeBERTa-base. These models were chosen for their superior ability to capture complex contextual nuances in natural language, significantly enhancing sensitivity to subtle language features associated with depression severity.

2.2. Research Goals

- Create a new method of depression detection from social media text using state-of-the-art pre-trained language models to incorporate subtle, context-dependent language cues for more substantial predictive power and more accurately characterize the severity of depressive symptoms.

- Evaluate contextual embeddings generated by sentence transformer models vs. traditional content-based encodings such as n-grams and their application relevance on labeling for finding the best-performing feature representations for depression detection.
- Investigate the capabilities of state-of-the-art models such as BERT, RoBERTa, and DeBERTa with novel encoding techniques to introduce optimal ways for estimating depression scores in user-generated text.

3. Proposed Methodology

This section provides a detailed overview of the datasets utilized in this study, the pre-processing techniques employed, and a comprehensive description of both the baseline methods and the proposed approach.

3.1. Dataset Details

We have utilized the data from the well-known social media website Reddit because people express their opinions, emotions, and behaviors on it. The data consist of posts with labels and classifications performed by clinical psychology evaluations (e.g., Beck Depression Inventory (BDI) or Manual of Mental Disorders (DSM-V), intended to evaluate depression severity. Reddit posts contain 24,000 instances of mild, moderate, or severe depression. Each post is chosen to be representative of a broad spectrum of depressive symptomatology, allowing for research across all levels of severity based on user language and content. This dataset provides a range of linguistic patterns illustrating the trajectory of depressed symptoms through authentic experiences and expressions associated with depression. Table 2 shows sample data with categories.

The lowest, mildest, and highest depression severity levels are based on clinical psychometric assessments, like the Beck Depression Inventory (BDI) and type of the Detailed and Statistical Manual of Mental Disorders (DSM-V). Posts marked as ‘Minimum’ have mild symptoms with low disruption in daily activities; in the space defined as ‘Moderate’, symptom burden, such as significant emotional and functional impairment, is assumed, while ‘Severe’ includes debilitating symptoms frequently coupled with thoughts of self-harm or suicidal ideation. These were labels that were assigned to them through annotation data by professionals in the field of mental health.

Table 2. Sample data from depressive posts.

No.	Post/Sentence	Severity Level
1	The five stages of grief are depression debilitating depression drunk Jim Morrison numbness exhaustion	Severe
2	My thoughts constantly ruin my mood	Moderate
3	Sometimes joking about depression is a coping mechanism for some people nobody is sad cause it's a trend.	Mild

3.2. Preprocessing and Analysis of Textual Data

Data preprocessing is an essential step for Natural Language Processing (NLP), which involves preparing raw text data to be cleaned, normalized, and ready for analysis. The process starts with eliminating non-relevant characters, such as punctuation symbols, memorable characters, and numeric digits, because they will not provide any semantic rationale to our text. Following that is text normalization, where you convert every character to lowercase. Hence, everything is uniform, and remove stop words, which are useless or common words like the, or, etc. Finally, the text is tokenized so that it can be split into single words or phrases (tokens). Next, we applied stemming or lemmatization to cluster

words of the same family together. It also includes spelling corrections and the removal of some residual noise, e.g., URLs, hashtags, or mentions in the case of social media data. The data preprocessing steps are illustrated in Figure 1.

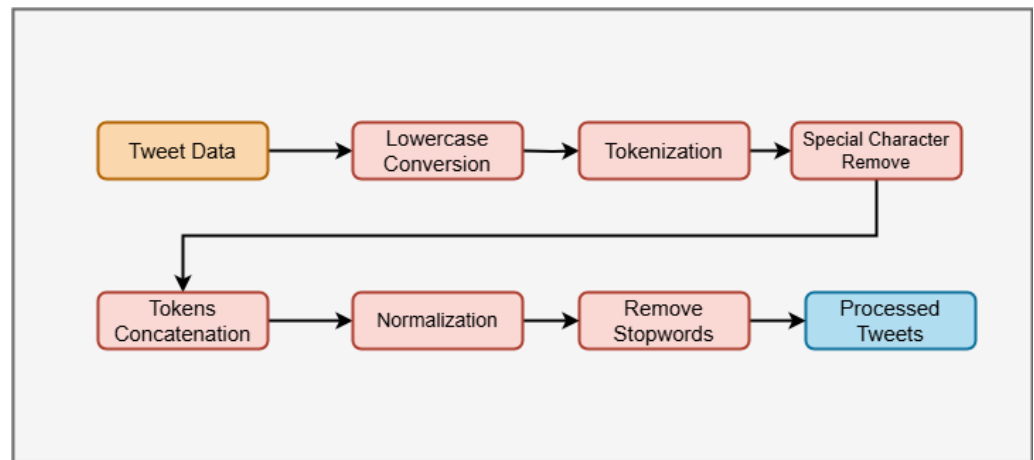


Figure 1. Data preprocessing Steps.

NLTK is a leading platform for building Python programs with human language data (text). It has a set of text-processing libraries for Classification, Tokenization, Stemming, Tagging, Parsing, and Semantic Reasoning. These NLTK filters have been visualized using a figure with an example sentence shown in Figure 2.

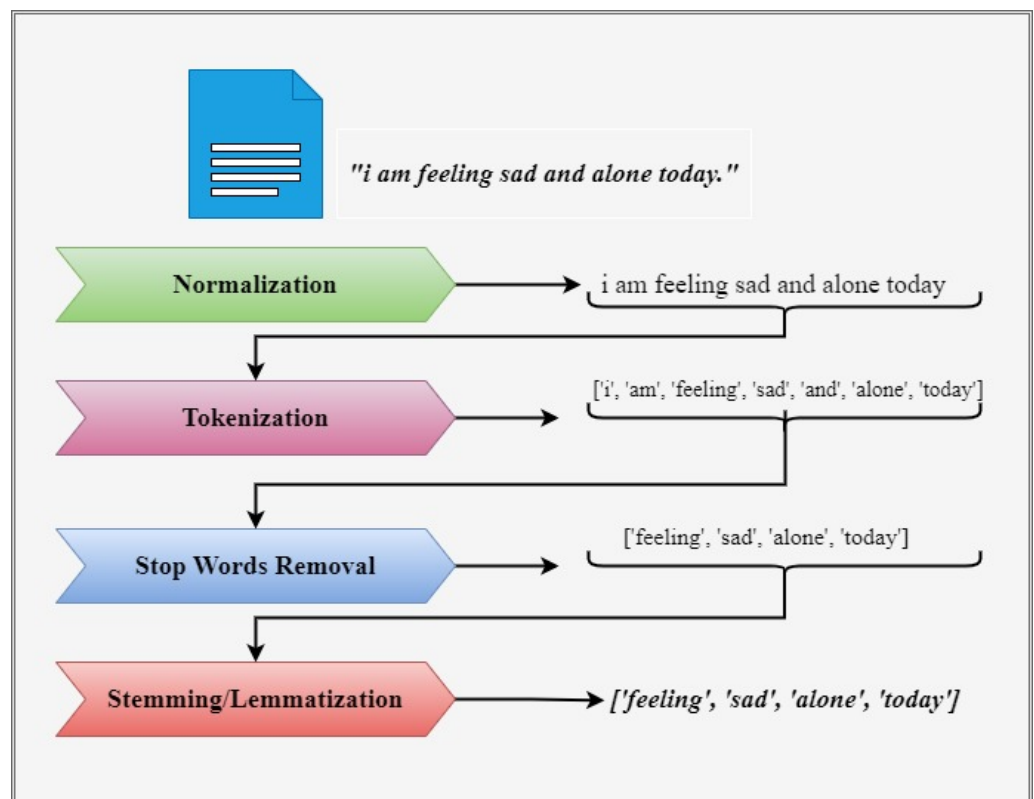


Figure 2. The filters of the NLTK library.

3.3. Approaches

Here we aim to demonstrate how the proposed methods enhance the precision of depression detection by leveraging a structured progression of approaches. Our study begins with content-based methods that utilize N-grams, followed by context-based methods

employing Sentence Transformers to capture deeper semantic representations. Building on these, we introduce a proposed end-to-end architecture based on advanced transformer models. This comprehensive framework provides a robust comparison between the baseline models and the novel methods, offering insights into strategies that have not been previously explored in experimental paradigms for depression detection. Detailed descriptions of each methodology and their contributions to improving detection are presented in the subsequent sections.

3.3.1. Classical Machine Learning-Based Approaches

This section explains the extraction of content-based features, such as n-grams, bi-grams, and tri-grams, and context-based feature extraction using sentence transformers for depression detection using classical machine learning models.

3.3.2. Content-Based Feature Extraction Using N-Grams

This study leverages content-based features for depression classification, specifically using N-grams after pre-processing text data to eliminate noises like special characters, stop words, and extra spaces as shown in Figure 3. N-grams are contiguous sequences of n-words that have been widely used for classification in the text process to capture syntactic and linguistic patterns. In the case of the paragraph above, uni-grams, bi-grams, and tri-grams are extracted from the text that was pre-processed in previous steps, which will ultimately lead to structured features capable of representing language patterns often associated with depression [43].

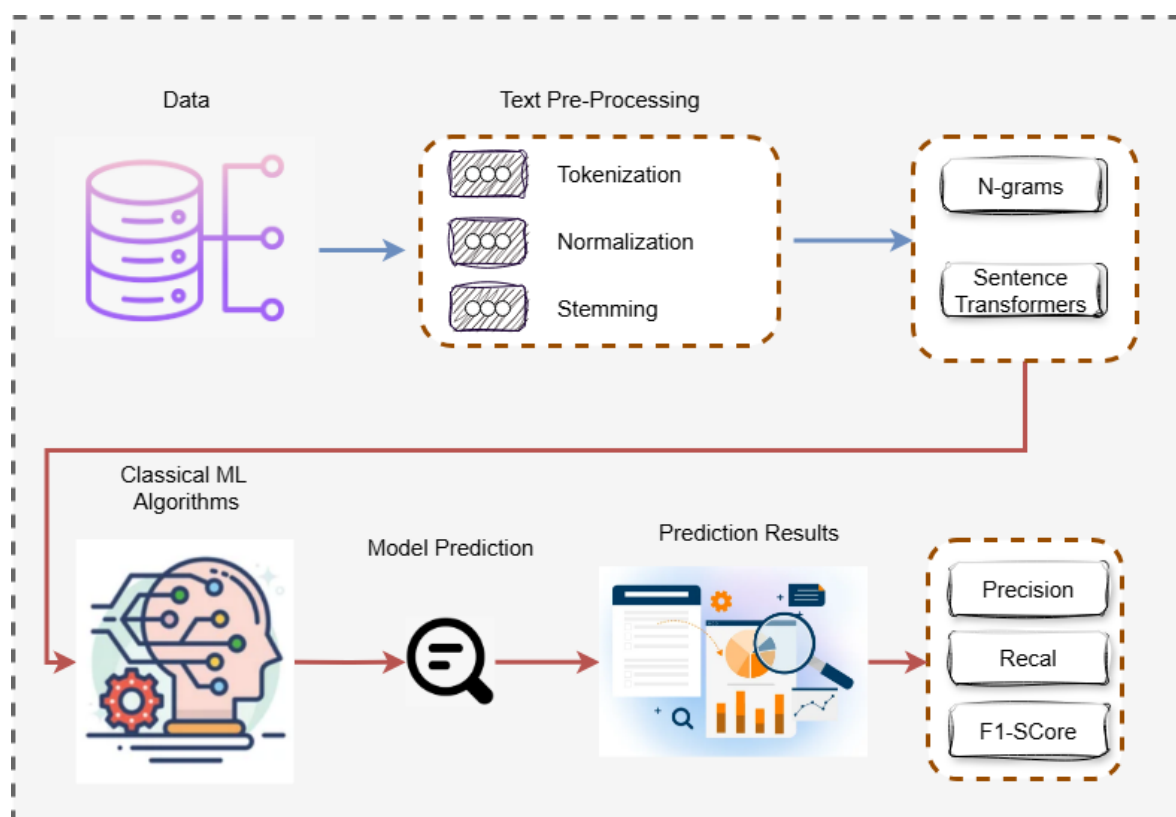


Figure 3. Classical machine learning-based approaches.

The use of N-grams as features was initially introduced for text classification and has since been effectively applied in detecting mental health conditions, including depression [44]. These features are transformed into numerical vectors using Term Frequency-Inverse Document Frequency (TF-IDF), a widely adopted weighting tech-

nique that ensures frequently used terms within depressive contexts are given higher importance [45].

N-gram-based approaches are instrumental in detecting depression linguistic markers like the increased usage of first-person pronouns (e.g., “I”, “me”) and negative-emotion words (“sad”, “hopeless”) [44]. The patterns captured from the corpus are now provided as inputs to classical machine learning models such as Random Forest (RF), Decision Tree (DT), Naive Bayes (NB), Gaussian Naive Bayes (GNB), Gradient Boosting Classifier (GBC), AdaBoost (AB), Multilayer Perceptron (MLP), K-Nearest Neighbor (KNN), Support Vector Machine (SVM) & Logistic Regression (LS). The models are trained and evaluated using precision, recall, and F1-score metrics. A detailed view of this approach can be seen using Figure 3. N-grams form a method specially designed to find the simple linguistic features of the text, such as sentence structure, based on the content. However, they cannot accurately describe the deeper context. In the following sections, we will overcome this limitation, where context-based approaches and end-to-end transformer architectures will be integrated to improve detection performance.

3.3.3. Context-Based Feature Extraction Using Sentence Transformers

After preprocessing, the context-based feature extraction using Sentence Transformers is employed to capture the semantic and contextual meaning of the text for depression detection. SBERT adapts the BERT model by introducing Siamese and triplet neural network structures, resulting in semantic sentence embeddings. SBERT is fast, multilingual (support for >100 languages), and works better than traditional approaches. For this study, eight pre-trained Sentence Transformer models are utilized: paraphrase-mpnet-base-v2, stsb-roberta-base-v2, all-MiniLM-L12-v2, paraphrase-xlm-r-multilingual-v1, all-distilroberta-v1, all-distilroberta-v1, all-MiniLM-L6-v2, distiluse-base-multilingual-cased-v2, and multi-qa-mpnet-base-dot-v. These models are fine-tuned for diverse NLP tasks and are particularly effective in capturing nuanced semantic features, making them well-suited for depression detection.

The embeddings generated by these models are passed to ten classical machine learning algorithms, including RF, DT, NB, GNB, GBC, AB, MLP, KNN, SVM, and LR, for classification. Model performance is evaluated using precision, recall, and F1-score metrics. A detailed view of this approach can be seen using Figure 3.

This approach significantly improves upon N-gram-based methods by capturing deeper semantic relationships within the text, crucial for identifying linguistic patterns associated with depression. The effectiveness of Sentence Transformers has been demonstrated in various NLP tasks, including text classification [46] semantic textual similarity [47], text reuse detection [48], and sentiment analysis [46], underscoring their potential in mental health research.

3.4. Proposed Transformer-Based Approaches

The proposed methodology employs advanced transformer-based models, including BERT-base-uncased, RoBERTa-base, and Microsoft/deberta-base, for depression detection. Unlike conventional approaches, which often involve separate steps to extract meaningful features for classification, these transformers can automatically learn representations for any task, thus allowing them to retain complex semantic and contextual details, which may be necessary for identifying linguistic features indicative of depression. The BERT-base-uncased processes the input through multiple layers of bidirectional attention to generate a set of comprehensive language representations. RoBERTa-base improves upon original BERT pretraining strategies (e.g., dynamic masking and more extended train period), enhancing contextual understanding and robustness. On the other hand, Microsoft/deberta-

base, which employs disentangled attention and relative position encoding, performs favorably in precision tasks due to its ability to model intricate associations between textual elements, thus being beneficial in identifying specific language related to depression [49].

These models eliminate the need for manual feature engineering, as they automatically learn task-specific features during fine-tuning. The proposed methodology for depression detection can also be viewed in Figure 4, which illustrates the systematic pipeline adopted in this study. The process begins with textual user post data collected as input, which undergoes essential preprocessing steps such as tokenization, normalization, and stemming to clean and standardize the text. After preprocessing, the dataset is split into training (70%), validation (10%), and testing (20%) subsets to ensure robust evaluation. Next, the data is passed through a feature extraction and classification pipeline, where advanced transformer-based models such as BERT-base-uncased, RoBERTa-base, and Microsoft/deberta-base are employed. These models are fine-tuned during this stage to extract meaningful linguistic and contextual features and classify the depression severity levels into three categories: Minimum, Moderate, or Severe.

In the last step, the predicted results from the transformers are validated by computing the classical metrics on precision, recall, and F1-score to confirm the effectiveness of the proposed methodology. The entire pipeline is described in Figure 4, which emphasizes the end-to-end process of detecting the severity of depression through preprocessing, fine-tuning, classification, and evaluation.

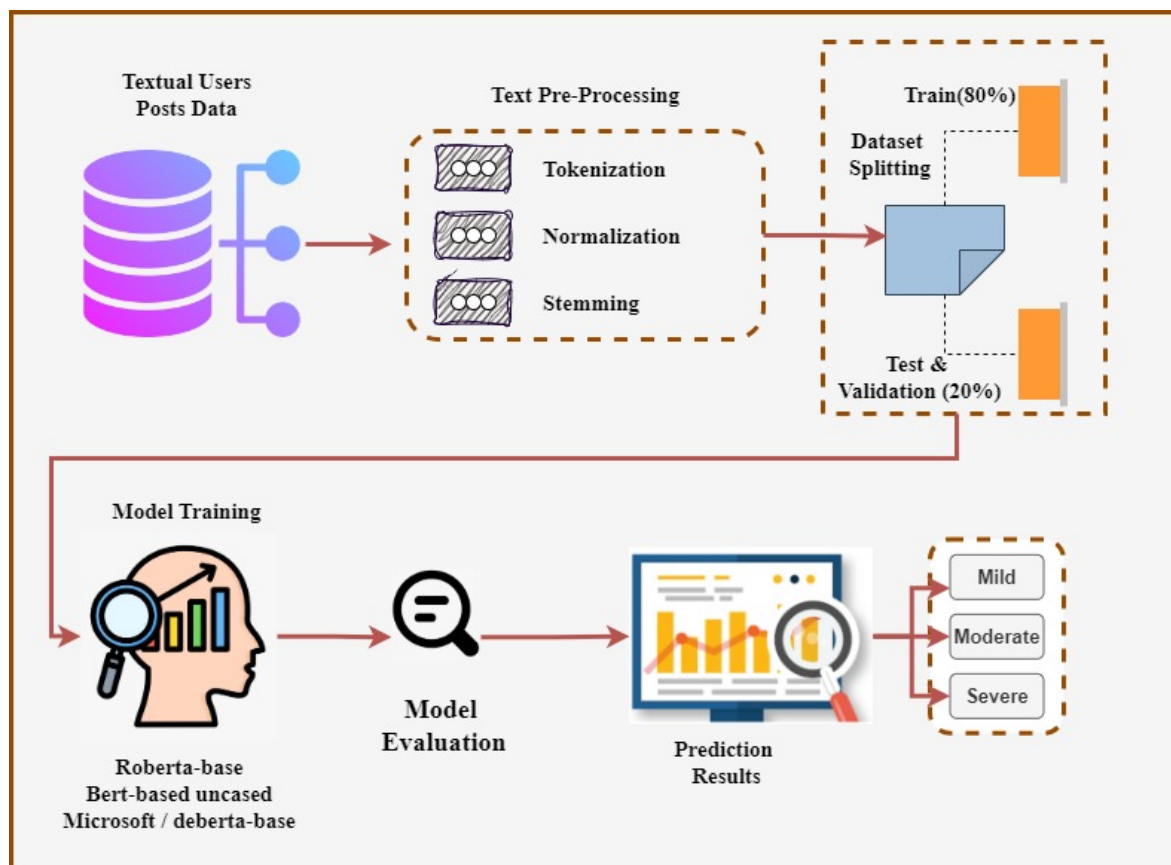


Figure 4. Proposed Transformer-based Approaches.

4. Experimental Setup and Results

In this section, we delve into the evaluation measures that we used to assess the performance of the proposed method and complete implementation details of the hyperpa-

rameters. The subsequent subsections comprehensively present the results obtained using the proposed method.

4.1. Evaluation Measure

Precision, recall, and F1 measures [44] as the average over depression dataset are this task's most frequently used evaluation measures.

- **Precision:** Precision (P) is defined as the share of True Positive (TP) predictions among all positive cases, which contains both True Positives and False Positives (FP):

$$P = TP / (TP + FP), \quad (1)$$

- **Recall:** Recall (R) gives the positive to predicted positive ratio:

$$R = TP / (TP + FN), \quad (2)$$

- **F1-measure:** The F1 measure is defined as the (F1) harmonic mean of two other numbers, Precision (P) and Recall R: In our study, we have used an average of F1-scores weighted:

$$F1 = 2 \times P \times R / (P + R). \quad (3)$$

4.2. Hyperparameter Setting

The proposed ST-based models were implemented in Python, utilizing NLTK. We applied the ReLU activation function and batch normalization for the experiments to enhance model stability, using the ADAMW optimizer with a learning rate of 0.0001. The training spanned 50 epochs with a batch size of 64. The dataset was divided into 70% for training, 20% for testing, and 10% for validation. The summary of hyperparameter settings is given in Table 3.

Table 3. Hyperparameter settings.

Hyperparameters	Value
Loss function	Cross Entropy
Learning rate	0.0001
Batch size	64
No. of epochs	50
Optimizer	ADAMW
The hidden layer activation function	ReLU
The output layer activation function	Softmax

4.3. Results Using Content-Based Method

Table 4 highlights the performance of N-gram models in depression detection. uni-grams (Exp 1.1–Exp 1.3) outperform other N-grams, achieving precision, recall, and F1-scores between 0.80 and 0.81, particularly with RF and MLP classifiers. Their simplicity effectively captures information without adding unnecessary complexity. In contrast, bi-grams show moderate performance with scores ranging from 0.61 to 0.67, offering slight improvements in contextual understanding but at the cost of increased complexity. Tri-grams perform the worst, with scores between 0.49 and 0.64, likely due to sparsity and noise. Overall, paired with RF or MLP, uni-grams provide the most reliable and balanced performance, confirming their suitability for content-based depression detection.

Table 4. Results using content based method.

Exp No.	N-gram	ST Models	ML Models	Precision	Recall	F1 Score
Exp 1.1	Uni-gram	Uni-gram 1000	RandomForestClassifier	0.80	0.81	0.81
Exp 1.2		Uni-gram 1500	MLPClassifier	0.81	0.81	0.81
Exp 1.3		Uni-gram 2000	MLPClassifier	0.80	0.80	0.80
Exp 1.4	Bi-gram	bi-gram 1000	MLPClassifier	0.61	0.61	0.61
Exp 1.5		bi-gram 1500	RandomForestClassifier	0.65	0.65	0.65
Exp 1.6		bi-gram 2000	MLPClassifier	0.67	0.67	0.67
Exp 1.7	Tri-gram	tri-gram 1000	MLPClassifier	0.49	0.49	0.49
Exp 1.8		tri-gram 1500	SVC	0.52	0.52	0.52
Exp 1.9		tri-gram 2000	MLPClassifier	0.64	0.54	0.54

4.4. Results Using Context-Based Method

Table 5 presents the performance of context-based methods for depression detection, utilizing ST models in combination with machine learning classifiers. Among the tested combinations, multi-qa-mpnet-base-dot-v1 (Exp 2.1) paired with XGB Classifier achieves the highest F1 score of 85.4, demonstrating its strong ability to capture semantic and contextual features critical for accurate classification. Similarly, Paraphrase-mpnet-base-v2 (Exp 2.6) with XGB Classifier delivers comparable results, with an F1 score of 85.5, showcasing its effectiveness in leveraging advanced sentence embeddings for this task. Other models, such as all-distilroberta-v1 (Exp 2.2) combined with MLP Classifier and distiluse-base-multilingual-cased-v2 (Exp 2.5) with Random Forest Classifier, also exhibit high performance with F1 scores exceeding 85.0, indicating their robustness in handling depression-related linguistic nuances. While slightly lower, all-MiniLM-L12-v2 (Exp 2.3) and Paraphrase-xlm-r-multilingual-v1 (2.8) demonstrate competitive performance, achieving F1 scores of 84.5 and 84.3, respectively. Notably, the XGB Classifier emerges as a consistent top performer across multiple sentence transformers, emphasizing its suitability for handling the extracted contextual features. The results highlight the superior classification capability of context-based methods, with sentence transformers effectively encoding semantic relationships, making them highly suitable for depression detection tasks. This performance underscores the advantage of combining advanced embeddings with robust classifiers.

Table 5. Results using context-based method.

Exp No.	ST Models	ML Models	Precision	Recall	F1 Score
Exp 2.1	multi-qa-mpnet-base-dot-v1	XGBClassifier	85.3	85.5	85.4
Exp 2.2	all-distilroberta-v1	MLP Classifier	85.3	85.2	85.2
Exp 2.3	all-MiniLM-L12-v2	KNeighbors Classifier	84.3	84.4	84.5
Exp 2.4	all-MiniLM-L6-v2	XGBClassifier	83.7	83.7	83.7
Exp 2.5	distiluse-base-multilingual-cased-v2	RandomForestClassifier	84.4	84.4	84.4
Exp 2.6	Paraphrase-mpnet-base-v2	XGBClassifier	85.3	85.8	85.5
Exp 2.7	Stsb-roberta-base-v2	XGBClassifier	83.4	85.5	84.5
Exp 2.8	Paraphrase-xlm-r-multilingual-v1	RandomForestClassifier	84.5	83.5	84.3

4.5. Results Using Proposed Transformer-Based Method

The proposed methodology was evaluated using three transformer-based models: RoBERTa-base, BERT-base-uncased, and Microsoft/DeBERTa-base for different scales of depression levels. Each model was trained and tested for various learning rates, the results shown in Table 6, demonstrating remarkable progress in assessing the severity of depression.

The RoBERTa-base performed the best at the learning rate equal to 0.000001. The Mild class achieved an excellent F1-score of 0.95 with a precision of 0.93 and a recall of 0.98. The f1-score of the model was 0.86 and 0.91 for the Moderate and Severe classes, respectively. The results demonstrate the model's power in distinguishing these fine-grained linguistic shifts associated with depression severity. RoBERTa-base, with an overall F1 score of 0.91, was a good choice for obtaining context-dependent features about depression-related text. By comparison, the BERT-base-uncased model was also competitive, particularly at a 0.000001 learning rate. Relative to other classes, the high F1 score (0.96) alongside the precision (0.95) and recall (0.97) indicates the reliability of the algorithm's ability to detect mild depression. The F1-score of the "Moderate" class was 0.85, while that of the "Severe" class was slightly lower at 0.80, suggesting we could better identify severe cases. Despite this, the model attained a final F1 score of 0.90, which shows that it can generalize well across all classes. The Microsoft/DeBERTa-base model was the best performing one, notably at a learning rate 1×10^{-5} . It was able to achieve a solid F1-score of 0.96 for the "Mild" class, with fixed precision (0.95) and recall (0.97). The model performed equally well in the Moderate and Severe classes, with F1 Scores 0.95 and 0.96, respectively. The overall F1 score of 0.90 attained by DeBERTa-base showed a significantly better capacity to capture complex contextual nuances, making it well-suited for ratifying different grades of depression severity.

Table 6. Results using proposed transformer-based approaches.

Exp No.	Transformers	Learning Rate	Class	Precision	Recall	F1 Score	Overall F1 Score
Exp 3.1	Roberta-base	0.00001	Mild	0.87	0.99	0.92	0.90
			Moderate	0.85	0.73	0.90	
			Severe	0.87	0.87	0.91	
Exp 3.2	Roberta-base	0.000001	Mild	0.93	0.98	0.95	0.91
			Moderate	0.89	0.84	0.86	
			Severe	0.90	0.91	0.91	
Exp 3.3	Bert-base-uncased	0.00001	Mild	0.96	0.96	0.96	0.89
			Moderate	0.83	0.81	0.92	
			Severe	0.85	0.86	0.91	
Exp 3.4	Bert-base-uncased	0.000001	Mild	0.95	0.97	0.96	0.90
			Moderate	0.84	0.86	0.85	
			Severe	0.90	0.86	0.80	
Exp 3.5	Microsoft/deberta-base	0.00001	Mild	0.95	0.97	0.96	0.90
			Moderate	0.86	0.89	0.95	
			Severe	0.85	0.90	0.89	
Exp 3.6	Microsoft/deberta-base	0.000001	Moderate	0.85	0.85	0.85	0.90
			Severe	0.91	0.88	0.89	

The RoBERTa-base model, evaluated with learning rates of 0.00001 and 0.000001, demonstrated robust performance across all depression severity classes. At 0.00001, the model achieved high accuracy for the "Mild" class with 1183 correct predictions but showed notable misclassifications in the "Moderate" and "Severe" classes, as illustrated in the confusion matrix in Figure 5. The training accuracy approached 100%, with validation accuracy stabilizing at 90%, though validation loss exhibited fluctuations, as shown in Figures 11 and 12.

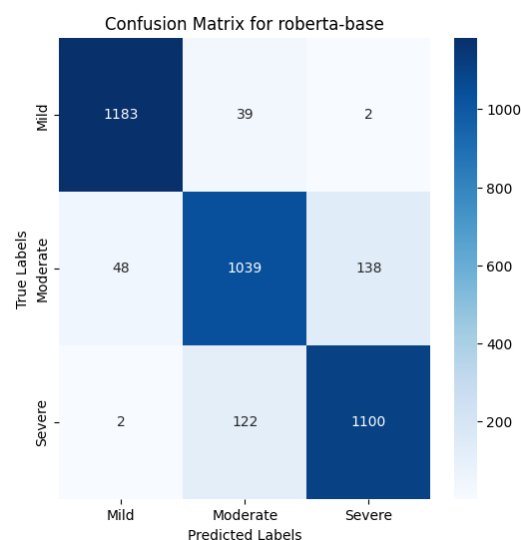


Figure 5. CM for RoBERTa-base model.

At 0.000001, the model showed improved classification performance, particularly for the “Mild” class, with 1198 correct predictions and fewer misclassifications in the “Moderate” and “Severe” classes, as observed in Figure 6. The training loss further stabilized, and validation accuracy remained consistent, as Figures 17 and 18 indicated, highlighting the model’s enhanced generalization. The finer learning rate of 0.000001 outperformed 0.00001, particularly in effectively reducing misclassifications and capturing contextual nuances.

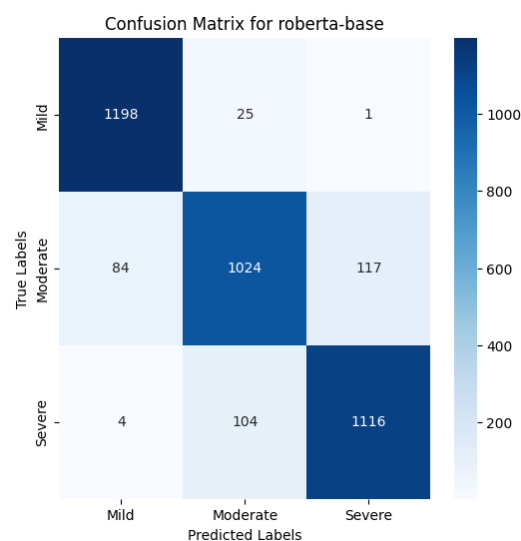


Figure 6. CM for RoBERTa-base model.

The BERT-base-uncased model demonstrated strong performance across all depression severity classes at learning rates of 0.00001 and 0.000001. At 0.00001, the model correctly classified 1183 “Mild” samples, 1030 “Moderate” samples, and 1074 “Severe” samples, as illustrated in Figure 7. Training loss stabilized at a low value, and validation accuracy remained consistent at 90%, as shown in Figures 13 and 14.

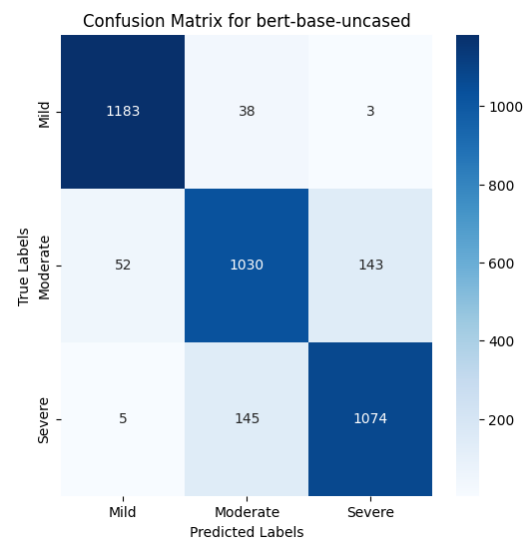


Figure 7. CM for bert-base-uncased model.

At 0.000001, the model showed improved performance, with 1191 correct “Mild” classifications and fewer misclassifications in the “Moderate” and “Severe” classes, as shown in Figure 8. The training loss remained stable, and validation accuracy showed similar consistency, as seen in Figures 19 and 20. Overall, the finer learning rate enhanced classification accuracy and reduced misclassifications, reinforcing the model’s ability to generalize effectively across depression severity levels.

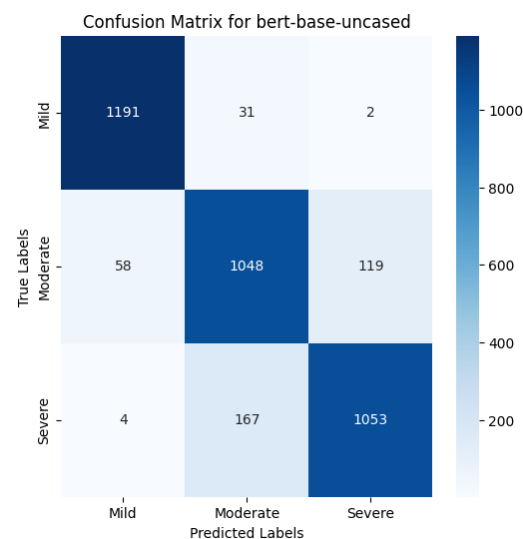


Figure 8. CM for bert-base-uncased model.

The Microsoft/DeBERTa-base model demonstrated strong performance at learning rates of 0.00001 and 0.000001. At 0.00001, the model correctly classified 1186 “Mild”, 1031 “Moderate”, and 1092 “Severe” samples, with minimal misclassifications, as shown in the confusion matrix in Figure 9. Training loss declined steadily and stabilized, while validation accuracy remained around 90%, as illustrated in Figures 15 and 16.

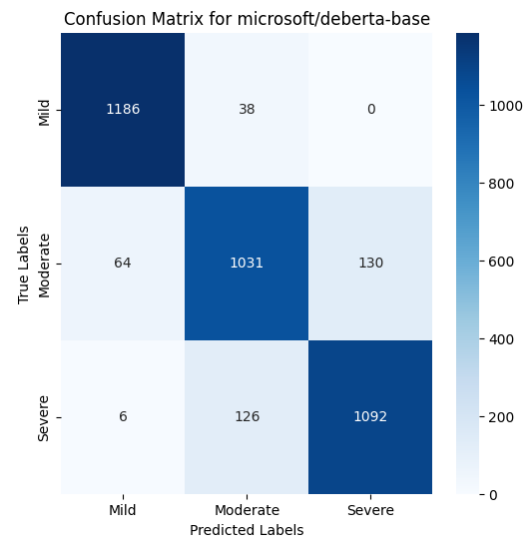


Figure 9. CM for microsoft/deberta-base model.

At 0.000001, the model improved slightly, with 1189 correct “Mild” classifications, 1044 for “Moderate”, and 1072 for “Severe”, reducing misclassifications, as seen in the confusion matrix in Figure 10. Training loss remained stable, and validation accuracy showed consistent results across epochs, as indicated in Figures 21 and 22. The finer learning rate enhanced classification accuracy and reinforced the model’s reliability in capturing depression severity nuances.

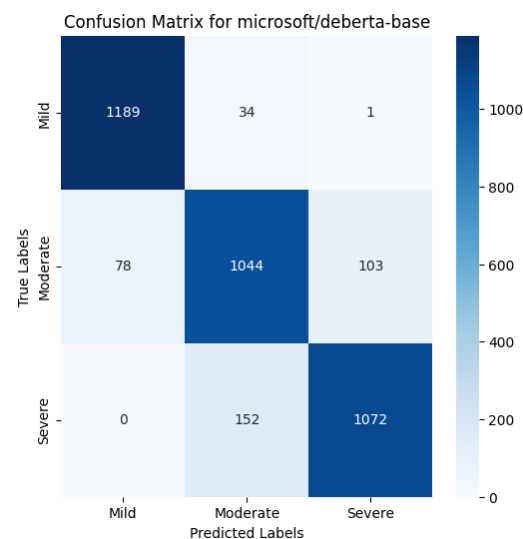


Figure 10. CM for microsoft/deberta-base model.

Finally, all three models had high precision and recall for all three classes, with the Mild Class significantly outperforming the other two classes. Among the models, Microsoft/DeBERTa-base emerged as the most effective, showcasing its capability to balance sensitivity and specificity across all depression severity levels.

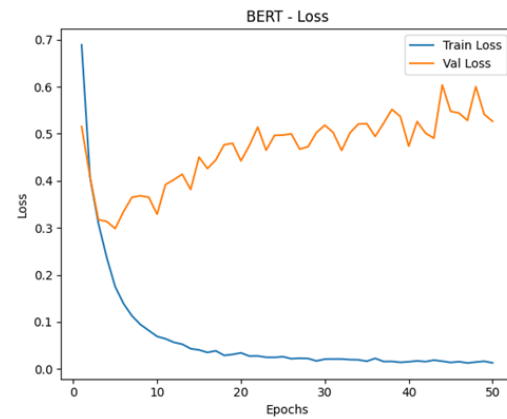


Figure 11. Loss of RoBERTa-base model.

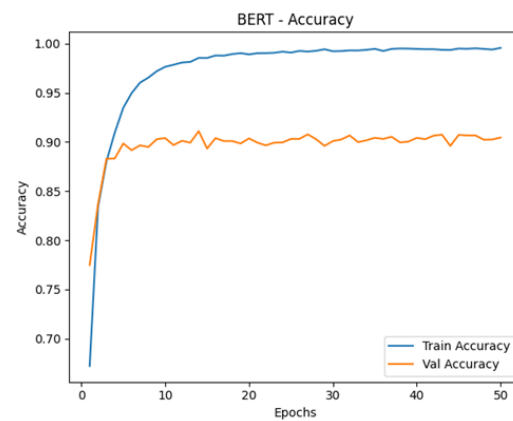


Figure 12. Accuracy of RoBERTa-base model.

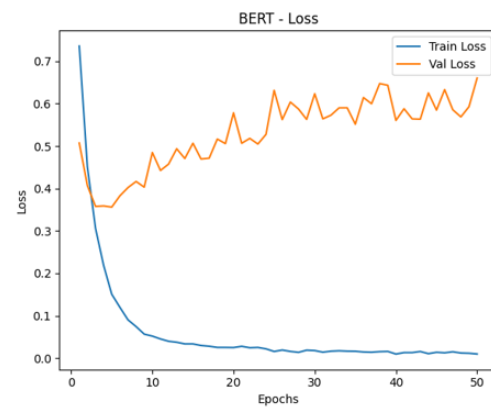


Figure 13. Loss of BERT-base-uncased model.

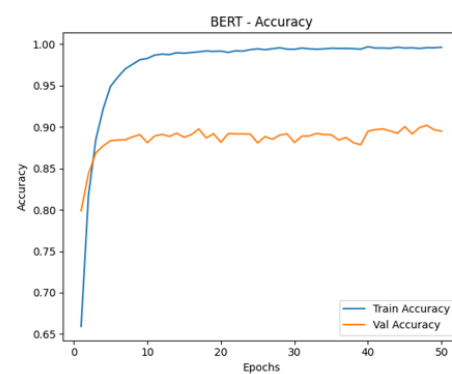


Figure 14. Accuracy of BERT-base-uncased model.

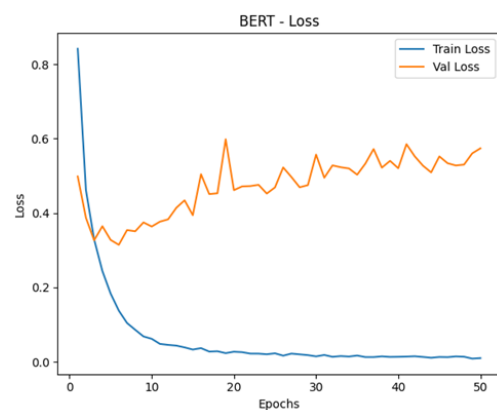


Figure 15. Loss of microsoft/deberta-base model.

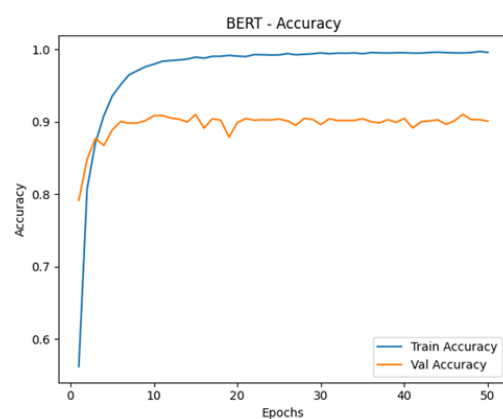


Figure 16. Accuracy of microsoft/deberta-base model.

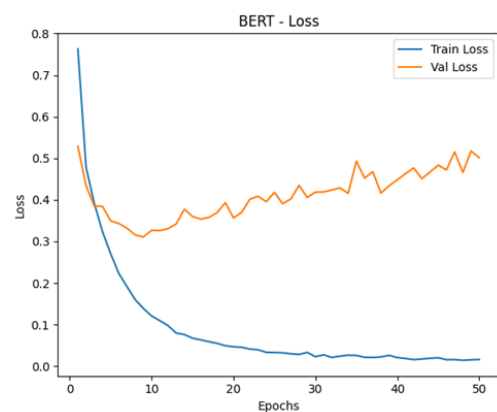


Figure 17. Loss of RoBERTa-base model.

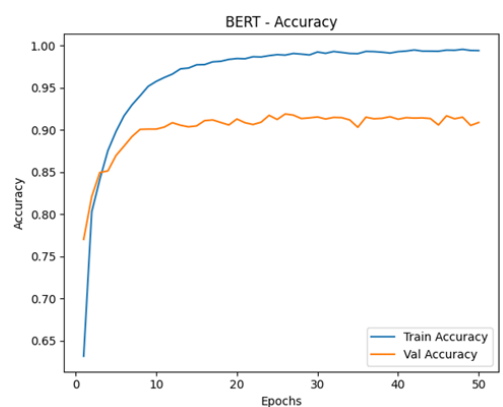


Figure 18. Accuracy of RoBERTa-base model.

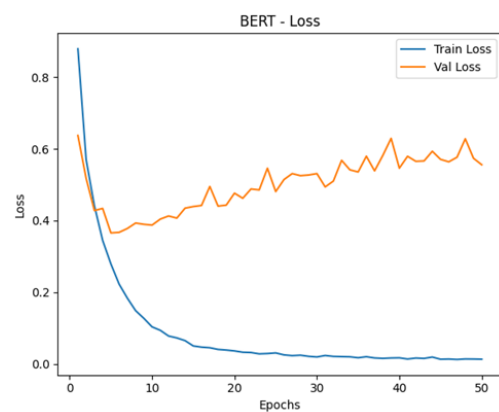


Figure 19. Loss of BERT-base-uncased model.

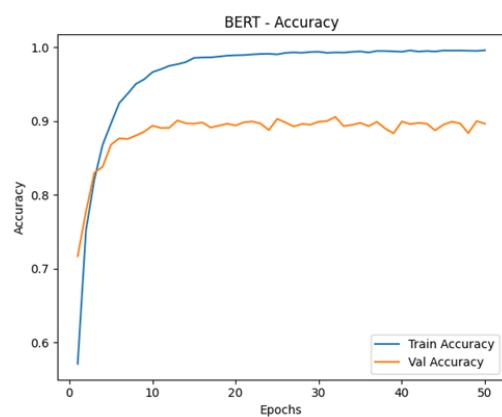


Figure 20. Accuracy of BERT-base-uncased model.

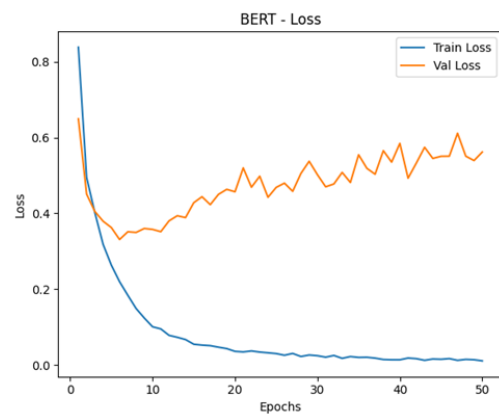


Figure 21. Loss of microsoft/deberta-base model.

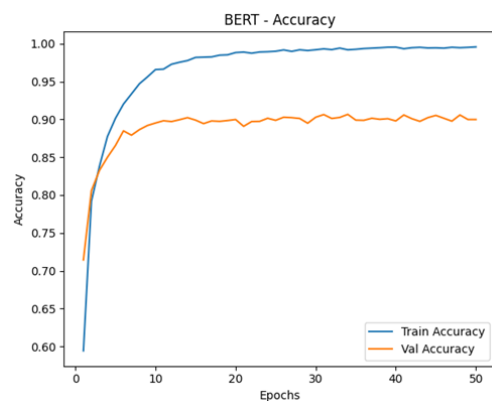


Figure 22. Accuracy of microsoft/deberta-base model.

5. Discussion

This study evaluates content-based, context-based, and transformer-based approaches for depression detection, highlighting the strengths and limitations of each. Content-based methods using N-grams, particularly uni-grams paired with RF or MLP classifiers, achieved reliable F1 scores of 0.80–0.81 by capturing basic linguistic patterns. However, their inability to model complex context and declining performance for bi-grams and tri-grams due to sparsity limits their effectiveness for nuanced text. Context-based methods leveraging Sentence Transformers significantly improved performance, with multi-qa-mpnet-base-dot-v1 and paraphrase-mpnet-base-v2 paired with XGB Classifiers achieving F1-scores of 85.4 and 85.5, respectively. While these methods effectively encode semantic relationships, they use static embeddings and, therefore, could struggle to adapt to slight nuances in the domain [50]. The transformer-based models (RoBERTa-base, BERT-base-uncased, and Microsoft/DeBERTa-base) proposed to resolve these limitations bring dynamic adaptability through fine-tuning and produce uniformly high precision, recall and F1 scores at all severity levels. For example, Microsoft/DeBERTa-base had robust classification with few misclassifications and stable generalization. Transformer-based models only managed to capture subtle context and semantics, enabling them to be the best-known answer regarding performance and eliminating previous models' flaws.

5.1. Comparison with SOTA Methods

This study evaluated transformer-based approaches for depression severity detection on Reddit data and compared their performance with existing methods on Twitter datasets, as seen in Table 7. Existing approaches, such as XGBoost [51], achieved limited performance, with an overall F1 score of 0.62, highlighting the limitations of traditional machine learning models in capturing linguistic context. Pre-trained transformer models [52] and dual-attention RoBERTa networks [53] demonstrated improved performance, achieving overall F1 scores of 0.88 and 0.87, respectively. While RoBERTa [54] achieved an F1 score of 0.90, its performance for Severe cases (0.89) was slightly lower than other methods.

Table 7. Comparison with SOTA Methods.

Ref.	Year	Dataset	Model	Class-Wise F1 Score			Overall F1 Score
				Mild	Moderate	Severe	
[28]	2023	Twitter	RoBERTa	0.96	0.86	0.89	0.90
[49]	2023	Twitter	XGBoost	0.63	0.70	0.53	0.62
[54]	2023	Twitter	Dual-Attention RoBERTa Network	0.97	0.87	0.89	0.87
[50]	2022	Twitter	Pre-trained Transformer Models	0.86	0.91	0.87	0.88
Proposed	2024	Reddit	Roberta-base	0.95	0.86	0.91	0.91

Among the proposed transformer-based models, RoBERTa-base outperformed the existing approaches, achieving the highest overall F1 score of 0.91. Notably, it gained strong classification performance across all classes, with F1 scores of 0.95 for Mild, 0.86 for Moderate, and 0.91 for Severe. This improvement is attributed to its fine-tuning capability, enabling it to adapt dynamically to nuanced contextual variations in Reddit data, outperforming models trained on Twitter datasets.

5.2. Case Study: Comparative Analysis of Methods

The advantages and disadvantages of different approaches used in this study are described in a case study. N-grams and other content-based methods are quick and efficient but suffer from inability to encode complex contexts. However, while context-based approaches, such as those utilizing Sentence Transformers, greatly enhance semantic com-

prehension, they are extremely compute-heavy and infeasible for use in smaller-scale applications. Alternatively, transformer-based models, e.g., RoBERTa and DeBERTa, combine the strengths of both approaches, dynamically capturing linguistic differences by considering the context of other tokens in a sentence, achieving very high precision and recall. However, these models require much more hardware resources and take longer to train. These observations highlight the balance between computational efficiency and classification accuracy.

In conclusion, the proposed transformer-based models demonstrated superior performance, particularly in balancing class-wise precision and recall, making them more effective for depression severity detection across diverse data sources. Their robustness and adaptability make them ideal for real-world applications in mental health monitoring.

6. Conclusions and Future Work

This study compared content-based, context-based, and transformer-based approaches for depression detection. While N-grams offered a reliable baseline, they lacked contextual depth. Sentence Transformers improved performance but were limited by fixed embeddings and adaptability. The proposed transformer-based models, particularly Microsoft/DeBERTa-base, outperformed both, delivering superior precision, recall, and accuracy through dynamic fine-tuning and robust contextual modeling, making them ideal for real-world applications. Future work should focus on integrating these models into clinical systems, expanding them to diverse datasets and languages, and exploring multimodal data for enhanced detection. Lightweight transformer models should also be developed for deployment in resource-constrained settings.

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